

ADVANCED REAL ANALYSIS: I. MEASURE THEORY

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ABSTRACT

These are some notes I made while reading through Folland's book [Fol99]. Real Analysis is a very difficult subject, especially at the graduate level. As such, I figured I'd make them public so other learners have more resources to study.

Hopefully, graduate students and those studying for PhD qualifying examinations in Analysis will find them useful.

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Please also email me if you discover any errors.

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"The trick with education is to teach people in such a way that they don't realise that they are learning until it's too late." - Harold Eugene Edgerton

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MOTIVATION

Since antiquity, mathematicians have been obsessed with finding the areas of regions bounded by curves and volumes in three-dimensional Euclidean space.

This longing led to the development of calculus, spearheaded by Newton and Leibniz, and later Riemann, resulting in a drastic increase in our understanding of the Universe.

The Riemann integral is sufficient for calculating the areas (or volumes) of regions bounded by "sufficiently nice" functions, enabling it to be taught to millions of high school students globally, but fails in more extreme cases.

Recall that we say a bounded function on a closed interval is Riemann integrable if its lower Riemann integral equals its upper Riemann integral¹

It is not hard to find a function that is not Riemann integrable, and perhaps the most famous example is the Dirichlet function, defined by

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

The non-integrability of this function is immediate from the density of $\mathbb{Q}$ in $\mathbb{R}$; any interval $[a, b]$ (where, of course, $a < b$), contains a real number and a rational number, so

$$\sup_{[a,b]} f = 1 \qquad \inf_{[a,b]} f = 0.$$

Consequently, for all partitions P of any closed interval $[c, d]$ in $\mathbb{R}$, $L(f, P, [c, d]) = 0$ and $U(f, P, [c, d]) = 1$. The upper and lower Riemann integrals of the Dirichlet function are, therefore,

$$U(f, [c, d]) = 1$$

and

$$L(f, [c, d]) = 0.$$

Citing page 9 of [Axl20], this example is disturbing because there are far fewer rational numbers than real numbers; recall that $\mathbb{Q}$ is countable, while $\mathbb{R}$ (and hence the set of irrationals) is uncountable²— intuitively, the integral of the Dirichlet function ought to be zero since the irrational contributions of "0" to the integral should drastically outweigh the rational contributions of "1".

¹The lower and upper Riemann integrals are defined by $L(f, [a, b]) = \sup_P L(f, P, [a, b])$, and $U(f, [a, b]) = \inf_P U(f, P, [a, b])$, for a bounded function $f : [a, b] \rightarrow \mathbb{R}$, respectively. Note the supremum is taken over all partitions P of the interval $[a, b]$.

²For an elegant proof of the uncountability of $\mathbb{R}$, see Cantor's Diagonalisation Argument.

Another issue is that the Dirichlet function (which I now restrict to $[0, 1]$ for simplicity) can be written as the pointwise limit of Riemann-integrable³ functions $f_k : [0, 1] \rightarrow \mathbb{R}$ defined by

$$f_k(x) = \begin{cases} 1, & x \in \{r_1, \dots, r_k\} \\ 0, & \text{otherwise,} \end{cases}$$

where $r_1, r_2, \dots$ is a sequence of each rational number⁴ (included only once) in $[0, 1]$. This shows the Riemann integral does not readily interchange with limits, as f is the pointwise limit of Riemann integrable functions, each being bounded by 1.

So is there another way? Well, to measure abstract volumes of sets, we would like to have a function μ that assigns a nonnegative real number, $\mu(E)$, to each subset $E \subset \mathbb{R}^n$ which satisfies countable additivity (which in turn implies finite additivity), meaning

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k)$$

for a (countably) disjoint sequence $\{E_k\}_{k=1}^{\infty}$ of subsets of $\mathbb{R}^n$.

Moreover, if E can be transformed into some other subset F through translations, rotations, and reflections, intuitively speaking, the size of E remains unchanged; the size of a ball's mirror image is the same as its own size. Hence, we would like $\mu(E) = \mu(F)$.

Last, it would make sense for $\mu(Q) = 1$, where Q denotes the unit cube in $\mathbb{R}^n$ —see page 19 of [Fol99].

Unfortunately, these definitions are inconsistent with one another (see page 20 of Folland). For an even more depressing example, one may look through 2.22 in [Axl20] to show that $\mathbb{R}$ (and hence $\mathbb{R}^n$) contains such pathological subsets that it is unreasonable for us to demand that μ be defined on *all* subsets of $\mathbb{R}^n$. This brings us to a discussion of σ -algebras.

1. SIGMA ALGEBRAS

Definition: σ -Algebras

Given a nonempty set X , we say that a set *of subsets* of X , $\mathcal{M}$, is a σ -algebra on X if we have the following:

- (1) $\emptyset \in \mathcal{M}$
- (2) If $E \in \mathcal{M}$, then $E^c \in \mathcal{M}$
- (3) If $\{E_k\}_{k=1}^{\infty}$ is a sequence of elements in $\mathcal{M}$, then the union $\bigcup_{k=1}^{\infty} E_k \in \mathcal{M}$.

Here is an example of a σ -algebra:

³The set of discontinuities of f_k for each $k \in \mathbb{N}$ is finite.

⁴We may do this as $\mathbb{Q}$ is countable; see Theorem 1.5.6(i) in [Abb15].

Proposition 1. *Let X be an uncountable set. Then the collection*

$$\mathcal{M} = \{E \subset X : E \text{ is countable or } E^c \text{ is countable}\}$$

is a σ -algebra on X .

Proof. $\emptyset \in \mathcal{M}$ and in particular, $\emptyset$ is countable, so that $\emptyset \in \mathcal{M}$.

Let $E \in \mathcal{M}$ be given. Now, if E is countable, $(E^c)^c = E$ is countable, so that $E^c \in \mathcal{M}$. If E^c is countable, that immediately satisfies our requirement.

To verify the last condition for a σ -algebra, let $\{E_n\}_{n=1}^\infty$ be a family of elements in $\mathcal{M}$. We proceed by cases:

Case 1: All the E_n 's are countable

Since the countable union of countable sets is countable⁵, then

$$\bigcup_{n=1}^\infty E_n \in \mathcal{M}$$

by definition.

Case 2: At least one E_n is only co-countable.

Then, since

$$E_n \subset \bigcup_{k=1}^\infty E_k,$$

taking complements yields

$$\left(\bigcup_{k=1}^\infty E_k \right)^c \subset E_n^c$$

so by assumption on E_n , $(\bigcup_{k=1}^\infty E_k)^c$ is a subset of a countable set, and is therefore countable. That is,

$$\bigcup_{k=1}^\infty E_k$$

is co-countable (and therefore in $\mathcal{M}$).

Cases 1 and 2 highlight that $\bigcup_{k=1}^\infty E_k$ is always contained in $\mathcal{M}$, so that by definition, $\mathcal{M}$ is a σ -algebra on X . $\square$

⁵See Theorem 1.5.8 (ii) in [Abb15]

We also note that σ -algebras are closed under countable intersection. Indeed, let $\{E_j\}_{j=1}^\infty$ be a sequence of sets in a σ -algebra $\mathcal{M}$ on a nonempty set X . Each complement $E_j^c \in \mathcal{M}$ by definition. By De Morgan's Law:

$$\left(\bigcap_{j=1}^\infty E_j \right)^c = \bigcup_{j=1}^\infty E_j^c \in \mathcal{M}$$

as σ -algebras are closed under countable union. Taking complements of the equation above shows that $\bigcap_{j=1}^\infty E_j \in \mathcal{M}$.

Theorem 1.1. *Let X be a set and $\mathcal{E} \subset \mathcal{P}(X)$ be a set of subsets of X . Then the intersection of all σ -algebras that contain $\mathcal{E}$ is a σ -algebra.*

Proof. Let $\mathcal{M}$ denote the intersection of all σ -algebras that contain $\mathcal{E}$. Each σ -algebra will contain $\emptyset$, which implies that $\emptyset \in \mathcal{M}$. Let $E \in \mathcal{M}$ be given. Then E must be an element of each σ -algebra on X that contains $\mathcal{E}$. Hence, E^c is an element of every σ -algebra on X that contains $\mathcal{E}$, which implies that $E^c \in \mathcal{M}$.

To show that $\mathcal{M}$ is closed under countable union, let $\{E_j\}_{j=1}^\infty$ be a sequence of subsets E_j of X contained in $\mathcal{M}$. Then each E_j is contained in all σ -algebras on X that contain $\mathcal{E}$, which implies that the union $\bigcup_{j=1}^\infty E_j$ is contained in all σ -algebras on X that contain $\mathcal{E}$. It follows that $\bigcup_{j=1}^\infty E_j \in \mathcal{M}$, as desired. $\square$

This result shows that there is a unique smallest⁶ σ -algebra on a set X that contains a given set $\mathcal{E}$ of subsets of X . We refer to this σ -algebra as the σ -algebra *generated* by $\mathcal{E}$, denoted by $\mathcal{M}(\mathcal{E})$. From this discussion, we can ascertain the following result (Lemma 1.1 in [Fol99]):

Theorem 1.2. *If $\mathcal{E} \subset \mathcal{M}(\mathcal{F})$, then $\mathcal{M}(\mathcal{E}) \subset \mathcal{M}(\mathcal{F})$.*

Proof. $\mathcal{M}(\mathcal{F})$ is a σ -algebra containing the set $\mathcal{E}$. $\mathcal{M}(\mathcal{E})$ was defined as the (unique) smallest σ -algebra that contains $\mathcal{E}$. Hence by definition, $\mathcal{M}(\mathcal{E}) \subset \mathcal{M}(\mathcal{F})$. $\square$

For a metric (or topological space) X , the σ -algebra generated by the set of open sets in X is known as the Borel σ -algebra, and denoted $\mathcal{B}_X$. Elements of $\mathcal{B}_X$ are called *Borel sets*.

The following result will be important when we learn about so-called measurable functions. That such families of sets can generate the Borel σ -algebra on the real line allows us to easily prove a criterion for the measurability of a function $f : X \rightarrow \mathbb{R}$, and this will be covered in my next article.

⁶For an intersection between two sets A and B , $A \cap B \subset A, B$. One can, therefore, regard the intersection as smaller than the two given sets A and B .

Theorem 1.3. *The Borel σ -algebra on $\mathbb{R}$, $\mathcal{B}_{\mathbb{R}}$, is generated by each of the following:*

- (1) *the open intervals $\mathcal{E}_1 = \{(a, b) : a < b\}$,*
- (2) *the closed intervals $\mathcal{E}_2 = \{[a, b] : a < b\}$,*
- (3) *the half-open (h) intervals $\mathcal{E}_3 = \{(a, b] : a < b\}$ or $\mathcal{E}_4 = \{[a, b) : a < b\}$,*
- (4) *the open rays $\mathcal{E}_5 = \{(a, \infty) : a \in \mathbb{R}\}$ or $\mathcal{E}_6 = \{(-\infty, a) : a \in \mathbb{R}\}$,*
- (5) *the closed rays $\mathcal{E}_7 = \{[a, \infty) : a \in \mathbb{R}\}$ or $\mathcal{E}_8 = \{(-\infty, a] : a \in \mathbb{R}\}$.*

Proof. See [Fol99], Proposition 1.2. □

Before defining what a measure is, we first discuss the concept of a product σ -algebra, which will help us develop integration theory in $\mathbb{R}^n$.

Definition: Product σ -algebra

Let $\{X_\alpha\}_{\alpha \in A}$ be an indexed collection of nonempty sets, X their Cartesian product $X = \prod_{\alpha \in A} X_\alpha$, and $\pi_\alpha : X \rightarrow X_\alpha$ the coordinate maps^a. If $\mathcal{M}_\alpha$ is a σ -algebra on X_α for each $\alpha \in A$, then we define the *product σ -algebra* on X as the σ -algebra generated by

$$\{\pi_\alpha^{-1}(E_\alpha) : E_\alpha \in \mathcal{M}_\alpha, \alpha \in A\}.$$

^aThe π_α are also referred to as projection operators.

We denote the product σ -algebra above with $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$. If, say, $A = \{1, \dots, n\}$, then we can also write $\bigotimes_{j=1}^n \mathcal{M}_j = \mathcal{M}_1 \otimes \dots \otimes \mathcal{M}_n$.

Theorem 1.4. *If A is countable, then $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is the σ -algebra generated by the sets*

$$\left\{ \prod_{\alpha \in A} E_\alpha : E_\alpha \in \mathcal{M}_\alpha \right\}.$$

Proof. Denote $\{\prod_{\alpha \in A} E_\alpha : E_\alpha \in \mathcal{M}_\alpha\}$ by $\mathcal{W}$. We will show that $\mathcal{M}(\mathcal{W}) = \mathcal{M}(\mathcal{E})$, where

$$\mathcal{E} = \{\pi_\alpha^{-1}(E_\alpha) : E_\alpha \in \mathcal{M}_\alpha, \alpha \in A\}.$$

To do this, note that if $E_\alpha \in \mathcal{M}_\alpha$, then we may write the preimage $\pi_\alpha^{-1}(E_\alpha) = \prod_{\beta \in A} E_\beta$, where

$$E_\beta = \begin{cases} X_\beta, & \beta \neq \alpha \\ E_\alpha, & \beta = \alpha. \end{cases}$$

This is because the preimage of E_α under π_α is defined as the set of $x \in X$ such that $x_\alpha \in E_\alpha$. Hence, since $X_\beta \in \mathcal{M}_\beta$ ⁷ and $E_\alpha \in \mathcal{M}_\alpha$, it follows that $\pi_\alpha^{-1}(E_\alpha) \in \mathcal{W}$. In particular, this shows that $\mathcal{E} \subset \mathcal{M}(\mathcal{W})$. Theorem 1.3 implies $\mathcal{M}(\mathcal{E}) \subset \mathcal{M}(\mathcal{W})$.

Conversely, note that we may write

$$\prod_{\alpha \in A} E_\alpha = \bigcap_{\alpha \in A} \pi_\alpha^{-1}(E_\alpha).$$

Indeed, $x \in \bigcap_{\alpha \in A} \pi_\alpha^{-1}(E_\alpha)$ if and only if $x \in \pi_\alpha^{-1}(E_\alpha)$ for all $\alpha \in A$, which is true if and only if $x_\alpha \in E_\alpha$ for all $\alpha \in A$, the case if and only if $x \in \prod_{\alpha \in A} E_\alpha$. Consequently since σ -algebras are closed under countable intersection, we have that $\mathcal{W} \subset \mathcal{M}(\mathcal{E})$. Again by Theorem 1.3, $\mathcal{M}(\mathcal{W}) \subset \mathcal{M}(\mathcal{E})$.

Combining the two inclusions gives the desired result of $\mathcal{M}(\mathcal{W}) = \mathcal{M}(\mathcal{E})$. $\square$

The σ -algebras $\mathcal{M}_\alpha$ in the definition of the product σ -algebra can be generated by families of sets $\mathcal{E}_\alpha$. The theorem⁸ below relates the generating collections $\mathcal{E}_\alpha$ of the $\mathcal{M}_\alpha$ to the generating collections of the product σ -algebra.

Theorem 1.5. *Suppose that $\mathcal{M}_\alpha$ is generated by $\mathcal{E}_\alpha$, $\alpha \in A$. Then the product $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is generated by*

$$\mathcal{F}_1 = \{\pi_\alpha^{-1}(E_\alpha) : E_\alpha \in \mathcal{E}_\alpha, \alpha \in A\}.$$

If we are also given that A is countable and $X_\alpha \in \mathcal{E}_\alpha$ for all α , then $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is generated by

$$\mathcal{F}_2 = \left\{ \prod_{\alpha \in A} E_\alpha : E_\alpha \in \mathcal{E}_\alpha \right\}.$$

Proof. $\mathcal{E}_\alpha \subset \mathcal{M}_\alpha$. Therefore,

$$\mathcal{F}_1 \subset \bigotimes_{\alpha \in A} \mathcal{M}_\alpha = \mathcal{M}(\{\pi_\alpha^{-1}(E_\alpha) : E_\alpha \in \mathcal{M}_\alpha, \alpha \in A\}).$$

Theorem 1.3 then implies $\mathcal{M}(\mathcal{F}_1) \subset \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$.

Conversely, for each α , it is easy to see that the set

$$\{E \subset X_\alpha : \pi_\alpha^{-1}(E) \in \mathcal{M}(\mathcal{F}_1)\}$$

is a σ -algebra on X_α that contains $\mathcal{E}_\alpha$ and hence $\mathcal{M}_\alpha$ (by Theorem 1.3). That is to say that for all $E \in \mathcal{M}_\alpha$, $\alpha \in A$, $\pi_\alpha^{-1}(E) \in \mathcal{M}(\mathcal{F}_1)$. Thus $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha \subset \mathcal{M}(\mathcal{F}_1)$. The other claim follows from the first as in the proof of Theorem 1.5. $\square$

⁷Recall that this must be the case since the $\mathcal{M}_\beta$ are σ -algebras on the spaces X_β .

⁸This is from Proposition 1.4 of [Fol99].

2. MEASURES

One of the biggest problems in Mathematics is finding a way to quantify size. Abstract *distances* can be understood in terms of metrics, which were then used to develop the theory of functions on metric spaces; see Chapter 19 of [Tao16] for a quick review.

We now turn to the issue of finding an abstract notion of *size*, or rather, *volume*.

Just as we defined a *metric* $d : X \times X \rightarrow [0, \infty]$ on a set X to satisfy certain properties⁹ that are in accordance with the familiar notion of distance in Euclidean space, we wish to define a generalised "*volume*" function $\mu : \mathcal{M} \rightarrow [0, \infty]$ ¹⁰ over a σ -algebra $\mathcal{M}$ which assigns an abstract volume $\mu(E)$ to a set $E \in \mathcal{M}$; recall that the σ -algebra is a family of sets.

Now, how do we generalise the size of a set? That is to say, given a σ -algebra $\mathcal{M}$, what properties should a function $\mu : \mathcal{M} \rightarrow [0, \infty]$ have?

For anything to be endowed with the responsibility of giving a generalised volume to a set, it must surely assign the value "0" to the empty set $\emptyset$, as it has no elements. Intuitively, if you imagine an empty water tank, then the amount of water inside, according to any reasonable scale of measurement, must be zero.

We therefore impose our first condition on μ : $\mu(\emptyset) = 0$.

What are some other desirable properties? Consider a sequence $\{E_k\}_{k=1}^{\infty}$ of disjoint sets in the σ -algebra. We ought to have the relation

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k).$$

To see why this must hold intuitively, consider cutting a cake into however many parts you like. Then certainly the sum of " μ " of all the slices would equal " μ " of the entire cake, which is equivalent to the *union* of all the pieces!

We therefore require μ to satisfy the two conditions discussed above, and call such functions a *measure*:

Definition: Measures

Let X a set equipped with a σ -algebra $\mathcal{M}$. A *measure* on $\mathcal{M}$ ^a is a function $\mu : \mathcal{M} \rightarrow [0, \infty]$ such that

- (1) $\mu(\emptyset) = 0$
- (2) If $\{E_k\}_{k=1}^{\infty}$ is a sequence of disjoint sets in $\mathcal{M}$, then
$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k).$$

^aIf $\mathcal{M}$ is clear from the context, we may simply say "on X ."

⁹Symmetry, $d(x, y) = 0$ if and only if $x = y$, and the Triangle Inequality.

¹⁰We relegate the image of μ to the nonnegative real line as it doesn't make sense for an object to possess negative volume or if you wish, size.

One remark: " $\mu(\bigcup_{k=1}^{\infty} E_k)$ " in condition (2) is well-defined because σ -algebras are closed under countable union.

(2) in the definition above immediately implies finite additivity, that is,

$$\mu\left(\bigcup_{k=1}^n E_k\right) = \sum_{k=1}^n \mu(E_k)$$

for a finite collection of disjoint sets $E_1, \dots, E_n \in \mathcal{M}$, because we may take $\{F_k\}_{k=1}^{\infty}$ to be

$$F_k = \begin{cases} E_k, & 1 \leq k \leq n, \\ \emptyset, & \text{otherwise.} \end{cases}$$

For X a set and $\mathcal{M} \subset \mathcal{P}(X)$ a σ -algebra, we call the pair $(X, \mathcal{M})$ a *measurable space*. Sets contained in $\mathcal{M}$ are referred to as *measurable*. If μ is a measure on $(X, \mathcal{M})$, then the triple $(X, \mathcal{M}, \mu)$ is called a *measure space*.

Functions $\mu : \mathcal{M} \rightarrow [0, \infty]$ that satisfy (1) and *finite* additivity (not necessarily (2)) are called *finitely additive* measures.

Here is an example of a measure.

Let X be an uncountable set, and let $\mathcal{M}$ be the σ -algebra of countable or co-countable sets.¹¹

Define the function $\mu : \mathcal{M} \rightarrow [0, \infty]$ by

$$\mu(E) = \begin{cases} 0, & E \text{ countable} \\ 1, & E \text{ co-countable,} \end{cases}$$

where $E \subset X$. Then μ constitutes a measure on $(X, \mathcal{M})$.

Proof: The empty set is countable, so $\mu(\emptyset) = 0$. To prove countable additivity, consider first a disjoint sequence $\{E_k\}_{k=1}^{\infty}$ of countable sets in X . The countable union of countable sets is countable, so

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = 0 = \sum_{k=1}^{\infty} \mu(E_k).$$

Now, if we had some $j \in \mathbb{N}$ such that E_j is co-countable, the union $\bigcup_{k=1}^{\infty} E_k$ is also co-countable (See Proposition 1 in Section 1). Moreover, disjointness implies that $E_k \subset E_j^c$ for all $k \neq j$, i.e., the E_k 's are subsets of a countable set for $k \neq j$, and are therefore countable. By definition of μ ,

¹¹See Proposition 1, Section 1.

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = 1, \quad \sum_{k=1}^{\infty} \mu(E_k) = \mu(E_j) = 1,$$

so, in general, we have shown that, for a disjoint sequence $\{E_k\}_{k=1}^{\infty}$,

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k),$$

that is, μ possesses countable additivity. ■

The Dirac Measure.

Let X be a nonempty set, and consider the σ -algebra $\mathcal{M} = \mathcal{P}(X)$ (the power set). Choose an $x_0 \in X$ and define the function $f : X \rightarrow [0, \infty]$ by

$$f(x) = \begin{cases} 1, & x = x_0 \\ 0, & x \neq x_0. \end{cases}$$

Then the map $\mu : \mathcal{M} \rightarrow [0, \infty]$ defined by $\mu(E) = \sum_{x \in E} f(x)$ is a measure on $\mathcal{M}$.

Proof:

$$\mu(\emptyset) = \sum_{x \in \emptyset} f(x) = 0.$$

For countable additivity, suppose we are given an arbitrary collection $\{E_k\}_{k=1}^{\infty}$ of disjoint sets contained in $\mathcal{M}$. If *none* of these sets contain x_0 , then

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = 0 = \sum_{k=1}^{\infty} \mu(E_k),$$

since $x_0 \notin \bigcup_{k=1}^{\infty} E_k$.

If one of these sets contains x_0 , then by disjointness none of the others contain x_0 . This implies

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = 1 = \sum_{k=1}^{\infty} \mu(E_k).$$

This proves countable additivity. ■

The Dirac Measure was named after the Nobel Prize-winning physicist Paul Dirac, who predicted the existence of the positron, later confirmed experimentally. He is also credited with laying the groundwork for Quantum Field Theory.

In general, a function $f : X \rightarrow [0, \infty]$ determines a measure μ by the sum

$$\mu(E) = \sum_{x \in E} f(x).$$

Indeed, for $E = \emptyset$, we have $\mu(\emptyset) = 0$, which can be easily deduced from the equation above. Countable additivity holds, because if $\{E_k\}_{k=1}^{\infty}$ is a collection of disjoint sets in $\mathcal{M}$,

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{x \in \bigcup_{k=1}^{\infty} E_k} f(x) = \sum_{k=1}^{\infty} \sum_{x \in E_k} f(x) = \sum_{k=1}^{\infty} \mu(E_k),$$

where the second equality holds by disjointness.

The next result shows that if $\mu_1, \dots, \mu_n$ are measures on a σ -algebra $\mathcal{M}$, and $a_1, \dots, a_n$ are nonnegative constants, then $\mu := a_1\mu_1 + \dots + a_n\mu_n$ is a measure on $\mathcal{M}$.

The Linear Combination of Measures is Also a Measure.

Suppose X is a set with a σ -algebra $\mathcal{M}$, and that $\mu_1, \dots, \mu_n$ are measures on $\mathcal{M}$. Then, for any nonnegative $a_1, \dots, a_n$, the function $\mu : \mathcal{M} \rightarrow [0, \infty]$ defined by

$$\mu := \sum_{k=1}^n a_k \mu_k$$

is a measure on $\mathcal{M}$.

Proof: Observe

$$\mu(\emptyset) = \sum_{k=1}^n a_k \mu_k(\emptyset) = 0.$$

This verifies the first property. For the second property, suppose $\{E_n\}_{n=1}^{\infty}$ is a sequence of disjoint sets in $\mathcal{M}$. Then

$$\begin{aligned} \mu\left(\bigcup_{n=1}^{\infty} E_n\right) &= \sum_{k=1}^n a_k \mu_k\left(\bigcup_{n=1}^{\infty} E_n\right) \\ &= \sum_{k=1}^n a_k \sum_{n=1}^{\infty} \mu_k(E_n) \\ &= \sum_{k=1}^n \sum_{n=1}^{\infty} a_k \mu_k(E_n). \end{aligned}$$

All terms in the double sum above are nonnegative by assumption, so we may interchange the order of summation:

$$\begin{aligned}\mu\left(\bigcup_{n=1}^{\infty} E_n\right) &= \sum_{n=1}^{\infty} \sum_{k=1}^n a_k \mu_k(E_n) \\ &= \sum_{n=1}^{\infty} \mu(E_n),\end{aligned}$$

by our definition of μ . ■

Measures have some useful properties, as outlined in this theorem.

Theorem 2.1 (Folland 1.8). *Let $(X, \mathcal{M}, \mu)$ be a measure space. Then,*

- (1) *(Monotonicity)*
If $E, F \in \mathcal{M}$ and $E \subset F$, then $\mu(E) \leq \mu(F)$.
- (2) *(Subadditivity)*
If $\{E_k\}_{k=1}^{\infty} \subset \mathcal{M}$, then $\mu(\bigcup_{k=1}^{\infty} E_k) \leq \sum_{k=1}^{\infty} \mu(E_k)$.
- (3) *(Continuity from below)*
If $\{E_k\}_{k=1}^{\infty} \subset \mathcal{M}$ and $E_1 \subset E_2 \subset \dots$, then $\mu(\bigcup_{k=1}^{\infty} E_k) = \lim_{k \rightarrow \infty} \mu(E_k)$.
- (4) *(Continuity from above)*
If $\{E_k\}_{k=1}^{\infty} \subset \mathcal{M}$ and $E_1 \supset E_2 \supset \dots$, and $\mu(E_1) < \infty$, then $\mu(\bigcap_{k=1}^{\infty} E_k) = \lim_{k \rightarrow \infty} \mu(E_k)$.

The proof of this result can be found on page 27 of [Fol99].

Before we proceed, a discussion on finitely additive measures.

Recall that a map $\mu : \mathcal{M} \rightarrow [0, \infty]$ that satisfies $\mu(\emptyset) = 0$ and

$$\mu\left(\bigcup_{j=1}^n E_j\right) = \sum_{j=1}^n \mu(E_j)$$

for any finite collection of disjoint sets $\{E_j\}_{j=1}^n$ in the σ -algebra $\mathcal{M}$ (but not necessarily countable additivity) is referred to as a *finitely additive* measure; it is a measure up to finite unions.

Let $\{E_j\}_{j=1}^{\infty} \in \mathcal{M}$ be a sequence of disjoint sets. We observe the inclusion

$$E_1 \subset E_1 \cup E_2 \subset E_1 \cup E_2 \cup E_3 \subset \dots$$

If we define the collection $\{F_k\}_{k=1}^{\infty}$ by

$$F_k = \bigcup_{j=1}^k E_j,$$

we notice that $F_1 \subset F_2 \subset \dots$, and $\bigcup_{k=1}^{\infty} F_k = \bigcup_{j=1}^{\infty} E_j$.

As μ has finite additivity,

$$\mu(F_k) = \sum_{j=1}^k \mu(E_j).$$

Now look at (3) in Theorem 2. Our discussion above implies that μ is a measure if it is continuous from below, because

$$\mu\left(\bigcup_{k=1}^{\infty} F_k\right) = \mu\left(\bigcup_{j=1}^{\infty} E_j\right) = \lim_{k \rightarrow \infty} \mu(F_k) = \lim_{k \rightarrow \infty} \sum_{j=1}^k \mu(E_j) = \sum_{j=1}^{\infty} \mu(E_j)$$

and μ therefore admits countable additivity.

Suppose $(X, \mathcal{M}, \mu)$ is a measure space, and consider a set E in the σ -algebra $\mathcal{M}$. If $\mu(E) = 0$, we say that E is a *null set*. We have the following important result:

Theorem 2.2. *Let $\{E_j\}_{j=1}^{\infty}$ be a sequence of null sets. Then the countable union*

$$\bigcup_{j=1}^{\infty} E_j$$

is also a null set.

Proof. This follows immediately from the subadditivity property of measures, as in Theorem 2 (2). Indeed, we have the constraint¹²

$$0 \leq \mu\left(\bigcup_{j=1}^{\infty} E_j\right) \leq \sum_{j=1}^{\infty} \mu(E_j) = 0,$$

which forces $\mu\left(\bigcup_{j=1}^{\infty} E_j\right) = 0$. □

Now is a good time to introduce the "almost-everywhere" terminology. Consider a measure space $(X, \mathcal{M}, \mu)$. If a statement about points x in X is true except for x in a null set, we say that it is true *almost everywhere* (which can be abbreviated as *a.e.*), or for *almost every* x . If we wish to be more specific, we can write μ -a.e.

For example, let $(X, \mathcal{M}, \mu)$ be a measure space, and consider two functions $f, g : X \rightarrow \mathbb{R}$. If $f(x) = g(x)$ for all $x \in X \setminus E$, where $E \subset X$ is such that¹³ $\mu(E) = 0$, then we say that $f = g$ μ -a.e.

¹²Why does the left-hand inequality hold?

¹³Obviously, we treat $E \in \mathcal{M}$.

3. ANALOGY TO FINITE-DIMENSIONAL VECTOR SPACES

The measure of two sets is analogous to the dimension of the sum of two finite-dimensional vector subspaces; first recall the familiar theorem from linear algebra¹⁴

Dimension of Two Subspaces.

Let V_1 and V_2 be subspaces of a finite-dimensional vector space. Then

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2).$$

If we view the dimension as the size of the vector space, then the formula above is reminiscent of how (long ago) we were taught to find the area of two intersecting shapes: You first find the sum of their two individual areas, and then subtract the area of the region where they overlap.

A similar result holds for measures:

Proposition 2. *If $(X, \mathcal{M}, \mu)$ is a measure space and $E, F \in \mathcal{M}$, then*

$$\mu(E \cup F) = \mu(E) + \mu(F) - \mu(E \cap F),$$

provided $\mu(E \cap F) < \infty$.

To prove this result, we need a lemma:

Lemma 1. Suppose $(X, \mathcal{M}, \mu)$ is a measure space and $E, F \in \mathcal{M}$ are such that $E \subset F$. Then, if $\mu(E) < \infty$, $\mu(F \setminus E) = \mu(F) - \mu(E)$.

Proof: As $F = E \cup (F \setminus E)$ and this union is disjoint, $\mu(F) = \mu(E) + \mu(F \setminus E)$, so since $\mu(E) < \infty$, we may subtract $\mu(E)$ from both sides to obtain

$$\mu(F \setminus E) = \mu(F) - \mu(E),$$

as desired. $\square$

Proof of Theorem 3: We may write

$$E \cup F = (E \setminus (E \cap F)) \cup (F \setminus (E \cap F)) \cup (E \cap F).$$

Since the union above is disjoint, finite additivity of the measure implies

$$\mu(E \cup F) = \mu(E \setminus (E \cap F)) + \mu(F \setminus (E \cap F)) + \mu(E \cap F).$$

The intersection $E \cap F$ is a subset of *both* E and F , so invoking Lemma 1,

$$\begin{aligned} \mu(E \cup F) &= (\mu(E) - \mu(E \cap F)) + (\mu(F) - \mu(E \cap F)) + \mu(E \cap F) \\ &= \mu(E) + \mu(F) - \mu(E \cap F) \end{aligned}$$

¹⁴For a proof, see 2.37 in [Axl24]

which is the desired result. ■

4. OUTER MEASURES

We are now ready to introduce the concept of an *outer measure*. In the definition below, please understand that an outer measure is not necessarily a measure.

Definition: Outer Measures

An *outer measure* on a nonempty set X is a function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ that satisfies the following.

- (1) $\mu^*(\emptyset) = 0$
- (2) If $A \subset B$, then $\mu^*(A) \leq \mu^*(B)$.
- (3) For a sequence $\{A_j\}_{j=1}^{\infty} \in X$, $\mu^*(\bigcup_{j=1}^{\infty} A_j) \leq \sum_{j=1}^{\infty} \mu^*(A_j)$

The third condition in the definition above is known as *countable subadditivity*.

After reading the definition above, it is reasonable to ask *how* we can construct outer measures. If we have a collection $\mathcal{E}$ of sets (which contains $\emptyset$ and X) with some notion of size; that is, we have a function $\rho : \mathcal{E} \rightarrow [0, \infty]$ with $\rho(\emptyset) = 0$, we may estimate the area of arbitrary sets $A \subset X$ by calculating the "smallest area" of covers of A . This is concisely stated in the theorem below, which can be found as Proposition 1.10 in Folland [Fol99].

Theorem 4.1. Let $\mathcal{E} \subset \mathcal{P}(X)$ and $\rho : \mathcal{E} \rightarrow [0, \infty]$ be such that $\emptyset \in \mathcal{E}$, $X \in \mathcal{E}$, and $\rho(\emptyset) = 0$. For any $A \subset X$, define a map $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ by

$$\mu^*(A) = \inf \left\{ \sum_{j=1}^{\infty} \rho(E_j) : E_j \in \mathcal{E}, A \subset \bigcup_{j=1}^{\infty} E_j \right\}.$$

Then μ^* is an outer measure on X .

I shall not prove this theorem here, for the proof can easily be found in Folland. However, two comments.

Remember that we wish to approximate the areas of arbitrary subsets of X using covers. We therefore need to ensure that there exists a cover of each A , whose members belong to $\mathcal{E}$. This can indeed be achieved, for by assumption, $X \in \mathcal{E}$, so we will always be able to cover any subset $A \subset X$ (though the approximation may perhaps not be "accurate").

The second thing I'd like to mention is regarding the proof of condition (2) in the definition of an outer measure. If $A \subset B$, then by hypothesis, any cover of B will also be a cover of A (a cover of A is not necessarily a cover of B). In particular, the set over which you take the infimum of in the definition of $\mu^*(A)$ includes the corresponding set in the definition of $\mu^*(B)$ —you take the infimum of a larger set that contains the smaller set, so it follows that $\mu^*(A) \leq \mu^*(B)$.

We now know how to construct outer measures. But we want to form measures since they are more "well-behaved"; for example, outer measures need not satisfy

countable additivity. The following definition is important to obtain measures from outer measures.

Definition: μ^* -Measurability

Suppose μ^* is an outer measure on a set X . We say that a subset $A \subset X$ is μ^* -measurable^a if, for all $E \subset X$,

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

^a μ^* -measurable sets are also referred to as Carathéodory measurable sets.

Let's ponder what this means geometrically. For a set A that is μ^* -measurable, consider any subset E of X . Our definition says that the "area" of the set E , $\mu^*(E)$ is equal to the *sum* of the area of the region contained in both E and A , $\mu^*(E \cap A)$, and the area of the region in E that is not contained in A , $\mu^*(E \cap A^c)$. That is, we can find the area of any arbitrary set E in the space X by calculating the area of the intersection between E and A , the area of the region that "solely belongs" to E (the "stuff" in E , outside A), and then adding the two values¹⁵. Basically, μ^* -measurable sets align perfectly with our intuition.

It is important to note that we may lessen the requirement in the definition above to

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c),$$

for all subsets E of X with $\mu^*(E) < \infty$. Indeed, because μ^* is an outer measure, we may write

$$\mu^*(E) = \mu^*((E \cap A) \cup (E \cap A^c)) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

for all sets A and E , so it suffices to prove the reverse inequality, which is immediate if $\mu^*(E) = \infty$.

We now come to an amazing result.

Theorem 4.2 (Carathéodory's Extension Theorem). *If μ^* is an outer measure on X , the collection $\mathcal{M}$ of μ^* -measurable sets is a σ -algebra, and the restriction of μ^* to $\mathcal{M}$ is a complete measure.*

The proof of this result is involved and can be found in [Fol99]. Carathéodory's Extension Theorem is powerful because we can extract a measure (which is complete, meaning its domain contains all subsets of null sets) from an outer measure.

We shall now discuss premeasures, but first, we need the definition of an *algebra*.

Definition: Algebras

¹⁵If you cannot visualise this, draw out a Venn diagram on a piece of paper.

Let X be a nonempty set. An *algebra* of sets on X is a nonempty collection $\mathcal{A} \subset \mathcal{P}(X)$ with the following properties.

- (1) $\emptyset \in \mathcal{A}$
- (2) (*Closed under complements*)
If $E \in \mathcal{A}$, then $E^c \in \mathcal{A}$.
- (3) (*Closed under finite union*)
If $E_1, \dots, E_n$ are elements of $\mathcal{A}$, then the union $\bigcup_{j=1}^n E_j \in \mathcal{A}$.

We have one more definition, which will allow us to induce outer measures on a nonempty set X .

Definition: Premeasures

Suppose $\mathcal{A} \subset \mathcal{P}(X)$ is an algebra. A *premeasure* μ_0 is a function $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ such that

- (1) $\mu_0(\emptyset) = 0$
- (2) If $\{A_j\}_{j=1}^\infty$ is a sequence of disjoint sets in $\mathcal{A}$ such that ^a $\bigcup_{j=1}^\infty A_j \in \mathcal{A}$, then $\mu_0\left(\bigcup_{j=1}^\infty A_j\right) = \sum_{j=1}^\infty \mu_0(A_j)$.

^aWe need this assumption because we did not require algebras to be closed under countable union.

Consider a premeasure μ_0 over an algebra $\mathcal{A}$. Note that $\emptyset \in \mathcal{A}$ and $X \in \mathcal{A}$ (because algebras are closed under finite unions and complementation, so as $\mathcal{A}$ is nonempty, we have some set $E \in \mathcal{A}$, so since $X = E \cup E^c$, the claim follows.) We also have $\mu_0(\emptyset) = 0$ and that μ_0 is nonnegative. Now, look at the conditions of Theorem 5.

The premeasure μ_0 satisfies our requirements on ρ , so in particular, a premeasure μ_0 over an algebra $\mathcal{A}$ will *induce* an outer measure μ^* on X by

$$\mu^*(E) = \inf \left\{ \sum_{j=1}^\infty \mu_0(A_j) : A_j \in \mathcal{A}, E \subset \bigcup_{j=1}^\infty A_j \right\}.$$

And it gets better!

Theorem 4.3. *If μ_0 is a premeasure on $\mathcal{A}$, consider the outer measure μ^* induced, as in the expression above. Then we have the following.*

- (1) $\mu^*|_{\mathcal{A}} = \mu_0$.
- (2) Every set in $\mathcal{A}$ is μ^* -measurable.

I refer readers to 1.13 in [Fol99] to see a proof, but notice how this theorem gives us a great deal of information as to *what* sets are μ^* -measurable.

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